Measuring The Fractal Properties of The Sierpiński Triangle

Educational paper on the properties of the Sierpiński triangle, including the measurement of the area and perimeter.

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Abstract

This educational mathematical study focuses on the unique fractal introduced by a Polish mathematician Wacław Sierpiński in 1915, the Sierpiński triangle. The main objective of this study is to guide you through the different properties of this fractal and to understand how this shape behaves in an infinite space. Another aim of this study is to inspire and educate an audience that may not be familiar with the beauty of the Sierpiński triangle and help develop another perspective on fractals in general. All of the claims that are presented in this study are simply derived from facts that can be deduced from this shape by using basic to intermediate mathematics.

Keywords: The Sierpiński triangle, fractal, limit of computation, iterative process, infinite series, fractal area, fractal perimeter

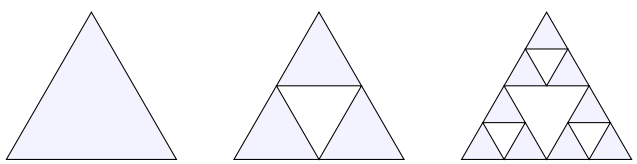
1 Introduction

The Sierpiński triangle is a unique geometric shape founded by a Polish mathematician Wacław Sierpiński in 1915, and is a triangular self-repeating shape, more commonly known as a fractal.

1.1 Key concepts and terminology

Before we can get started, we first need to address the key concepts and terminology that is going going to be mentioned throughout this study.

The Sierpiński triangle is a fractal constructed by an infinite series of smaller triangles that fit together in a pattern. A whole triangle is divided into four smaller ones, with the middle one not belonging to the whole shape. For every smaller triangle created, this process is infinitely repeated; therefore, continually smaller triangles are being subtracted from the whole shape.



(a) 1st iteration (b) 2nd iteration (c) 3rd iteration

Figure 1: The Sierpiński triangle throughout iterations

The fractal's area is always starting at A_0 units^2 and from there the area is each iteration getting smaller by $\frac{3}{4}$ of its previous area.

The triangle side length always starts at a units of length, and from there the side length is halved every iteration.

2 Fundamental values

This section of the educational paper focuses more on the theoretical side of things, on the subject matter of fundamental and arbitrary values of this fractal. Also, we will try to get a glimpse of the feasibility of computing these values and the processes behind all of the necessary calculations.

2.1 Fractal dimension

Every mathematical geometric shape holds a specific fractal dimension property to its form. Any geometric shape can be split up into a number of smaller, yet self-similar parts. The shape split by this process has a set of two attributes, the scaling factor and the number of copies. The scaling factor is a property that tells us how many smaller parts can be fit in a side of the original shape; and the number of copies is just the number of smaller copies of the original shape produced by splitting the shape up. We can compute the shape's fractal dimension through this formula:

$$\begin{aligned} s^d &= n \\ d &= \log_s(n) \\ d &= \frac{\log(n)}{\log(s)} \end{aligned} \tag{1}$$

where:

$d =$ the fractal dimension;

$s =$ the scaling factor, $s \in \mathbb{N}$;

$n =$ the number of copies, $n \in \mathbb{N} \setminus \{1\}$.

So, let's take for an example a square that we are going to split up into 9 equal smaller parts. The scaling factor s of the new smaller parts is equal to 3, since the side of the small part fits 3 times into the side of the big square. Furthermore, the number of copies n will be 9. From these values, we can assemble the fractal dimension's equation:

$$d = \log_3(9) = 2 \tag{2}$$

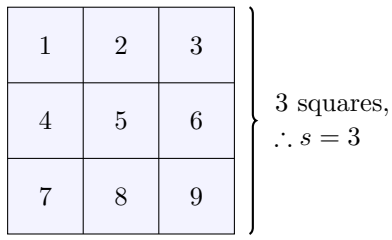


Figure 2: Fractal dimension of a square

Now for computing the fractal dimension of the Sierpiński Triangle itself. The scaling factor s of this fractal's smaller parts is 2, since only 2 small triangles can be fit into the side length of the fractal. Additionally, the total number of copies n is going to be 3. From discovering these values, we can substitute to the fractal dimension equation:

$$d = \log_2(3) \approx 1.585 \tag{3}$$

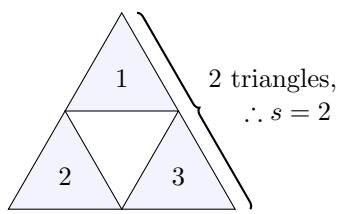


Figure 3: Fractal dimension of the Sierpiński Triangle

From this calculation, we can see that the Sierpiński triangle has no ordinary dimension number. It's fractal dimension is not a number that we could put onto a integer number line, since its too complex — it's approximately 1.585D. This is a another strange, yet fascinating feature of fractals, and this fractal specifically.

2.2 Feasibility of computations

Before trying to calculate other properties of this fractal, we must know if it is even possible to get a glimpse of its properties represented as finite values. To

answer this question, first, we must tear this shape apart to its core. This shape seems complex, but it is, in reality, constructed only from small triangles. This fact alone should point us in the right direction.

Axiom 1 *If we can tear apart a shape to make it simpler with each iteration, it should be possible to compute the area and the perimeter of this shape.*

The properties of this triangular fractal can be easily calculated and can also be logically derived from the facts that we have gathered about this shape. The compounded area of the Sierpiński triangle must have a finite value, since it is being stored within a 2D space that does not alter its size at any given point. Hence, the whole area is a finite number, despite the fact that the process through which we can obtain this value is infinite. Similarly, the perimeter is also stored in a 2D, non-altering space, but with one major difference. The perimeter gets longer with every iteration, unlike the area, which gets smaller due to the triangle division at every iteration.

3 Area computation at nth iteration

Before trying to compute the area at any given iteration, we first need to understand the process by which this value is constructed. This process, like previously said, is an infinite series of the same operations but executed on smaller and smaller parts of the fractal. Thus, we must account for all the possible n iterations from 0 to infinity.

3.1 Proof 1: Area reduction

The main triangle will get divided into 4 equal parts, with exactly 1 triangle being removed from the total area. You can notice that at every iteration the area is always reduced by a $\frac{1}{4}$th of the triangle's area; therefore, the area is always $\frac{3}{4}$ of the previous triangle's area. Now, we can model an equation that will express the relationship between n and the area.

The area at each step takes $\frac{3}{4}$ and multiplies it by itself n times. So, for simplicity of notation, we can raise it to the nth power. To also count in the initial triangle's area, we can plug in A_0 and multiply it by the expression:

$$A_n = \left(\frac{3}{4}\right)^n \cdot A_0 \tag{4}$$

where:

A_n = the computed area at nth iteration;

A_0 = the initial fractal area;

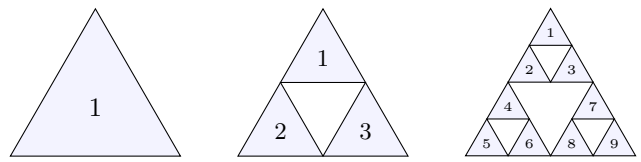
n = the current iteration, $n \in \mathbb{N}_0$.

3.2 Proof 2: Counting triangles

We can also use an alternative approach that will subsequently lead us to the same conclusion. This approach can appear to be more intuitive or comprehensive to the readers, since we are working with more arbitrary values than we were with the previous method.

In the last approach, we tried to reduce the area each iteration by a $\frac{1}{4}$th of

the triangle's area. In the second approach, we will try adding up all the small triangles' areas that belong to the shape. You can notice that the number of triangles inside the shape is always increased by a factor of 3. This means that the total number of triangles at nth iteration is always 3^n.



(a) 3^0 triangles (b) 3^1 triangles (c) 3^2 triangles

Figure 4: Number of triangles at nth iteration

Once we have established the number of triangles, we need to compute the area that each triangle is going to take up. The whole area of a triangle at the nth iteration is always divided into four equal parts. So, we can say that the area of one of those 4 triangles is equal to $\frac{1}{4}$ of its area.

Depending on the current iteration, the partial area will decrease as the iteration property increases. At every iteration, the area of the 3 remaining triangles is getting smaller by a $\frac{1}{4}$th. This means that the area of the small triangles is $\frac{1}{4^n}$ multiplied by the initial triangle's area.

We can now simply combine those two facts into an equation. If we know that every nth iteration has 3^n triangles and that every triangle has an area of $\frac{1}{4^n}$, that means that the area at any given iteration is:

$$\begin{aligned} A_n &= \frac{3^n}{4^n} \cdot A_0 \\ &= \left(\frac{3}{4}\right)^n \cdot A_0 \end{aligned} \tag{5}$$

As you can see, we have come to the same formula like we did in the previous equation (4). We can take this as a confirmation that this formula is correct and that no matter which approach we choose to take, the result will remain the same.

3.3 Proof 3: Visual representation

There exists a third way to compute the area at the nth iteration of the Sierpiński triangle; it is also the last alternative that we will cover in this paper. This alternative is more visual than the others, and its core depends on the rearrangement of the triangles that make up the whole fractal.

First, we need to lay down all the triangles next to each other that are present within a certain iteration. The first triangle is going to sit with its base down. We are going to lay the second triangle next to the first one with its base up, it might look like we are pushing the triangle down to the empty spot in the fractal. This process is repeated until we have run out of available triangles to lay in. We can visualize the second iteration in the following way:

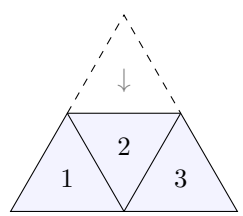


Figure 5: Rearrangement of triangles

Now, when we have laid down all the triangles that are available at the nth iteration, we can start rearranging the triangles into a new shape that can be more easily calculated. Take the last triangle and split it by the height of the equilateral triangle into two equal parts. Leave the first half of the triangle at the end of the series, but move the second half to the very beginning and flip it by the horizontal axis.

Notice that the area of the new shape did not change and is still equal to the previous series of laid-down triangles. However, one thing that did change was the shape. The new shape is a rectangle created by the triangle series and can now be more easily visualized. We can express the relationship between the

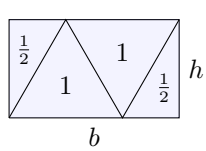


Figure 6: Triangles in a rectangular pattern

fractal area and the rectangle's properties in the following way:

$$A_n = b_n \cdot h_n \tag{6}$$

Two variables that still remain to be figured out are the base and the height of the new rectangle. Let us start with the base of the rectangle. You can notice that when all the triangles are laid down to fit in the rectangle, half of them have their bases down, and half of them have their bases up. This means that the length of the rectangle's base must equal the number of triangles in the desired iteration divided by 2, therefore $\frac{3^n}{2}$.

We must also take into account the side length of the triangles that make up the whole fractal; we can call the zeroth iteration's triangle side length a. We know that the side length of the first iteration's triangle is $\frac{1}{2}a$, so if we were to get the ratio between them, we would see that the side lengths keep getting smaller by the factor of $\frac{1}{2}$. This means that the side length of a triangle at nth iteration is equal to $\frac{1}{2^n}a$ or $\frac{a}{2^n}$.

Now, we just have to multiply the number of the triangle bases that make up the whole fractal with the size of the triangle's side lengths:

$$b_n = \left(\frac{3^n}{2}\right)\left(\frac{a}{2^n}\right) \tag{7}$$

Once we have established the base of the rectangle, we can move on to the height of the rectangle. You can notice that the rectangle's height is the same as the height of the triangles that are present in the rectangle. So, first, we need to determine the height of the equilateral triangles that make up the fractal. From a short calculation, we can say that a triangle with side length a has height $\frac{\sqrt{3}}{2}a$. This height, which we previously calculated, must also be scaled down by the side length of the fractal's triangles $\frac{a}{2^n}$.

Once again, we just have to multiply the height of the triangles with the size of the triangle's side lengths:

$$h_n = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{a}{2^n}\right) \tag{8}$$

Now, once the height and the base of the new-formed rectangle are settled, we can multiply them together to get the area, like that:

$$\begin{aligned} A_n &= \left(\frac{3^n}{2}\right)\left(\frac{a}{2^n}\right) \cdot \left(\frac{\sqrt{3}}{2}\right)\left(\frac{a}{2^n}\right) \\ &= \frac{3^n\sqrt{3}a^2}{4^{n+1}} \end{aligned} \tag{9}$$

3.4 Example

We have previously explored how to derive the formula for the computation of the area at the nth iteration. Now, we will take a look at a basic example of how to use this formula. For example, let's say that we want to compute the area of this fractal at the 3rd iteration, with the initial triangle's area being 10 units. We need to make use of the formula we came up with earlier:

$$\begin{aligned} A_{10} &= \left(\frac{3}{4}\right)^3 \cdot 10 \\ &\approx 4.219 \text{ units}^2 \end{aligned} \tag{10}$$

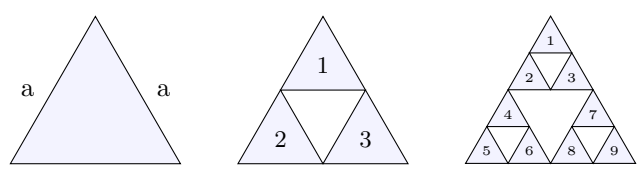
4 Perimeter computation at nth iteration

Before we can get started, we first need to take into account the properties on which this computation depends. In order to compute this, we need to find the total number of individual triangles that together compose the whole fractal and the lengths of the triangles.

4.1 Proof

First, we compute the total number of triangles that make up the fractal. You can notice that with each iteration, the total number of triangles is multiplied by a factor of 3. This means that the total number of triangles at any given iteration is 3^n. Once the total number of triangles is settled, we have to calculate the total number of sides that the triangles together contain. This means taking the total number of triangles at the nth iteration and multiplying it by a factor of 3, since each triangle has 3 sides; therefore, the total number of triangle sides is 3^{n+1}.

Now, we need to take into account that the triangle lengths are getting smaller with each iteration. We know that the side length of the first iteration's triangle is $\frac{1}{2}a$, so if we were to get the ratio between them, we would see that the side lengths keep getting smaller by the factor of $\frac{1}{2}$. This means that the side length of a triangle at nth iteration is equal to $\frac{1}{2^n}a$.



(a) 3^0 triangles (b) 3^1 triangles (c) 3^2 triangles

Figure 7: Number of triangles at nth iteration

The final formula comes down to the two properties, the number of triangle sides and the shrinking factor, when combined together in a product:

$$\begin{aligned} P_n &= 3^{n+1} \cdot \frac{1}{2^n} a \\ &= \frac{3^{n+1}}{2^n} \cdot a \end{aligned} \tag{11}$$

4.2 Example

We have previously explored how to derive the formula for the computation of the perimeter at the nth iteration. Now, we will take a look at a basic example of how to use this formula. For example, let's say that we want to compute the perimeter of this fractal at the 3rd iteration, with the initial triangle's side length being 10 units. We need to make use of the formula we came up with earlier:

$$\begin{aligned} P_{10} &= \frac{3^{3+1}}{2^3} \cdot 10 \\ &\approx 101.25 \text{ units} \end{aligned} \tag{12}$$

5 Approaching the limits of computation

We can now compute the precise area and perimeter properties at a certain iteration. Those values can help us when we are reckoning with finite nth iteration's values, but when we take into account the infinite side of things, those equations will not be able to provide us the solutions we desire anymore.

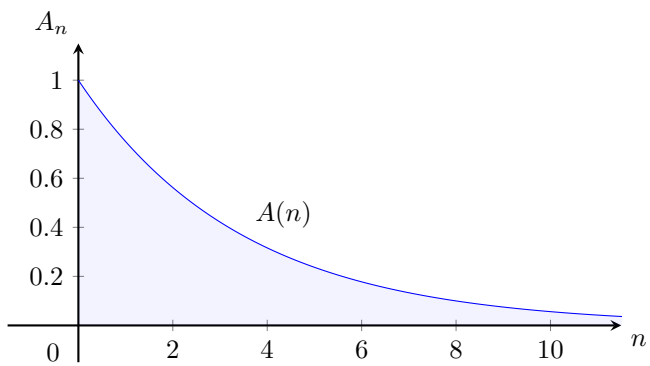
This is the moment when we need to introduce the limits of these computational functions. Testing the area and perimeter functions will help us solve these properties in an infinite space. We will be able to approach the limits by simulating extreme input to these functions.

5.1 Approaching the area limit

We have previously looked into the process of computing the area, from which we have received the rule for the calculation. Now, we can get the gist of that function, plug it into the limit L_a, and see to what extent it approaches. The gist of the area function is equal to $\left(\frac{3}{4}\right)^n$. Now, we need to plug it into the limit with respect to the variable n going to infinity:

$$L_a = \lim_{n \to \infty} \left(\frac{3}{4}\right)^n \tag{13}$$

Notice that no matter how high the variable n goes in the examined limit, the function does not increase in value. So, we can determine that the function is monotonically non-increasing, meaning that the area will get infinitely smaller with each iteration. We can depict this in a graph $A(n) = \left(\frac{3}{4}\right)^n$ and see how the output A_n will respond over iterations.



(14)

The constructed graph, with respect to both A_n and n, only further proves that the area will be getting infinitely smaller and will continue to get closer to 0. There is another way to prove this: notice that the denominator approaches infinity quicker than the numerator since $4^n > 3^n$; meaning that the whole fraction will keep getting smaller in result. Now, we can say for certain that the limit of the function will approach the value of 0.

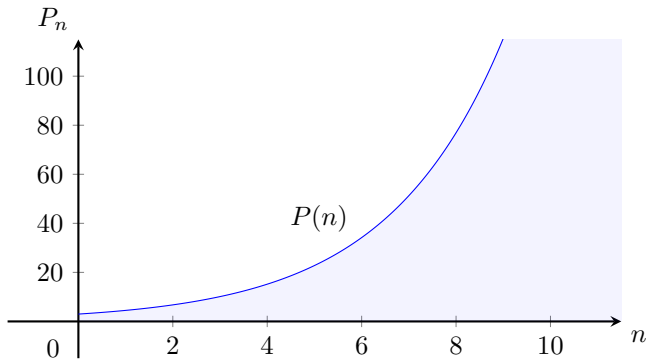
$$L_a = 0 \tag{15}$$

5.2 Approaching the perimeter limit

We have also discovered the rule for computing the precise perimeter of this shape at nth iteration. Now, we can plug the gist of that rule to the limit L_p abs see what extent it approaches. The gist of the function computing the perimeter is $\frac{3^{n+1}}{2^n}$, and we are going to plug it into the limit with respect to n going to infinity:

$$L_p = \lim_{n\to\infty}\left(\frac{3^{n+1}}{2^n}\right) \tag{16}$$

Notice that the output of this function is getting only bigger with the variable n. So, we can determine that this function is monotonically increasing, meaning that the perimeter will get infinitely larger with each iteration. We can depict this in a graph $P(n) = \frac{3^{n+1}}{2^n}$ and see how the output P_n will respond over iterations.



(17)

The constructed graph, with respect to both P_n and n, only further proves that the perimeter will be getting infinitely larger and will continuously get closer to infinity. There is another way to prove this: notice that the numerator approaches quicker than the denominator since $3^{n+1} > 2^n$; meaning that the whole fraction will keep getting greater in value. Now, we can say for certain that the limit of the function will approach infinity.

$$L_p = \infty \tag{18}$$

6 Interesting facts

6.1 Geometric series

An interesting happens when you try to use the sum of a geometric series on the Sierpiński triangle, more specifically on all of the area iterations throughout this shape. For the sake of this interesting fact, we can create an infinite set containing all of the area iterations including the 0th one:

$$A = \{A_0, A_1, A_2, A_3, \ldots, A_n\} = \{A_n \mid n \in \mathbb{N}_0\} \tag{19}$$

We can use the formula $\frac{a_1}{1-r}$ to calculate the sum of this geometric series, where a_1 is the first term of the series and r is the common ratio between the terms. The first term of the series A will be A_0. We are not going to use any specific values, since we want to generalize make the computation. The common ratio r is just the multiplication factor between the terms. In this case it is $\frac{3}{4}$, which can be derived it from the area function $A(n) = \left(\frac{3}{4}\right)^n$. The common ratio must also fulfill the condition $|r| < 1$, which this common ratio satisfies. Now, we just need to substitute these variables into the formula:

$$S_A = \sum_{n=0}^{\infty} A_n = \frac{A_0}{1 - \frac{3}{4}} = \frac{A_0}{\frac{1}{4}} = 4A_0 \tag{20}$$

This is interesting because the sum of all the area iterations of this shape is just 4 times the initial area of the triangle.

6.2 Sum

$$\begin{aligned}
&\sum_{n=1}^{\infty} \frac{3^{n-1}}{4^n} \cdot A_0 \\
&\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n \cdot \frac{1}{3} \cdot A_0 \\
&\frac{A_0}{3} \cdot \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n \\
&\sum_{n=1}^{\infty} x^n = \frac{a}{1-r}, \text{ for } |r| < 1
\end{aligned} \tag{21}$$

6.3 Pascal's triangle

$$
\begin{array}{ccccccccccccccc}
 & & & & & & & 1 & & & & & & & \\
 & & & & & & 1 & & 1 & & & & & & \\
 & & & & & 1 & & 2 & & 1 & & & & & \\
 & & & & 1 & & 3 & & 3 & & 1 & & & & \\
 & & & 1 & & 4 & & 6 & & 4 & & 1 & & & \\
 & & 1 & & 5 & & 10 & & 10 & & 5 & & 1 & & \\
 & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 & \\
1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1
\end{array}
$$

Figure 8: Pascal's triangle

7 Conclusion